Comparing Emergent Social Structures in RL and LLM-based Agents

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Abstract

Within ABMs, both RL agents and LLM-based agents have been shown to produce emergent social structures. No direct comparison of emergent social structures exists when the two agent architectures play the same coordination game. This thesis asks whether differences exist and what they are by comparing RL agents with LLM-based agents across prompt conditions.

1. Research Context

Research in emergent cooperation has increasingly focused on artificial agents, most prominently Reinforcement Learning (RL) agents and Large-Language Model (LLM)-based agents. RL agents make decisions based on their policies pretrained on reward-optimization through iterated interaction, while LLM-based agents act through token prediction based on large-scale pretraining on human language.

Despite these fundamentally different architectures, both agent classes have been shown to produce emergent social structures. This thesis addresses the question whether these architectures produce different patterns of social emergence in the same coordination environment.

2. Research Methods

To study emergent social phenomena, bottom-up agent-based models are commonly employed, following the traditions of Schelling and Epstein & Axtell. In this thesis, a minimal coordination game is designed to provide sufficient possibilities for emergent structures to be produced, without forcing outcomes.

The methodological focus is comparative as identical environments and interaction rules are used to analyze how different agent architecture produce emergent social structures.

2.1. Experimental Design

While it is impossible to isolate all architectural differences between RL and LLM-based agents, the experiment aims to minimize the confounding differences by fixing the environment, available information and optimization objective for both agents.

Experiments are run under three conditions, shown in Table 1.

Condition	Architecture	Information	Purpose
RL (LSTM)	Reward optimizing	Minimal state	Baseline
LLM-Comparison	Token prediction	Same as RL	Comparative condition
LLM-Control	Token prediction	Same as RL	Prompt variation

Table 1: Experimental conditions for comparing emergent social structure.

Where LLMs have memory through their context window, RL agents are essentially Markovian. By introducing an LSTM-state to the RL architecture, the agent can use the full information to optimize its policies during training and inform decision making during runtime.

The main comparison of the RL agent will be with a minimally prompted LLM agent, receiving the same information within a prompt. Additionally, they have a goal statement that asks them to optimize their resources.

Lastly, through varying the prompting of the LLM-based agent, the robustness of the emergent social structure is tested. Possible variations include tone, decoration, language and explicit instructions.

2.2. Game Design

Playing agents pursue the goal of optimizing their resources. Every round, agents simultaneously choose any action from Table 2. There is complete information: all resources and interaction history is available to all.

Action	Target	Effect
Invest	Self	costs V, generates I
Invest	Other	costs V, generates I
Arm	Self	costs D, shortly multiplies M
Arm	Other	costs D, shortly multiplies M
Attack	Other	Resource relocation

Table 2: Action space for agents in the coordination game.

Conflicts are resolved probabilistically based on the combined resources of the main combatants and active arm-other-interactions. The winner takes percentage T of the loser's resources. All combatants pay conflict costs C.

The game is played for N rounds (unknown to the players), with complete information including a history window enabling agents to track previous interactions.

2.3. Measurements

Emergent social structure is measured through network- and distribution-based metrics computed over interaction histories.

- **Coalition Structure**: Communities detected in directed interaction networks derived from investment and arm actions. Metrics include number of communities, average community size, and temporal stability.
- **Interaction Patterns**: Frequencies of cooperative and attack-actions and retaliation probabilities.

- **Hierarchy**: Inequality in resource distribution measured via the Gini coefficient over time.

3. Significance of Research

Research on artificial agents has progressed on two relatively isolated paths. Literature has shown that both RL and LLM-based agent architectures can produce emergent social structures, but no direct comparison has been made. The results of my thesis provide researchers with information on how their choice of agent architecture affects results. Additionally, the thesis is a methodological contribution as the first direct comparative study in a controlled environment on emergent social structures.

4. Bibliography

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5. Tentative Timeline

Milestone	Date
Thesis start	February 1, 2026
Literature review & environment setup	February 2026
Experiments	March – April 2026
Analysis & writing	May – July 2026
Thesis submission	July 15, 2026
Defence	Late July 2026